

Le Hoang Viet

Website | GitHub | Scholar

Ho Chi Minh City, Vietnam

lehoangviet2k@gmail.com | +(84) 36-3314-108

EDUCATION

Ho Chi Minh University of Technology – Vietnam National University

B.Eng. in Automation & Control Engineering

August 2018 – November 2023

Ho Chi Minh City, Vietnam

EXPERIENCE

AI Engineer

Monorevo Inc.

April 2025 – Present

Ho Chi Minh City, Vietnam

- Applied a VLM with Structured Output Prompting and Chain-of-Thought reasoning to engineering drawing analysis, replacing PaddleOCR and achieving a 35% accuracy improvement
- Fused MatMul-Activation layers in the LLM serving pipeline, delivering 16% faster inference at production scale
- Implemented the Rocchio algorithm for human-based relevance feedback in 2D/3D similarity search

Computer Vision Engineer

Ftech Co. Ltd

November 2023 – April 2025

Da Nang, Vietnam

- Fine-tuned SegFormer-B0 with motion-aware augmentation (+13% mIoU) and deployed it as a C++ OBS plug-in
- GPU-ported OpenCV's `guidedFilter` and alpha-matting in PyTorch, eliminating CPU bottlenecks on high-resolution video

AI Engineer

Bosch Global Software Vietnam

May 2023 – November 2023

Ho Chi Minh City, Vietnam

- Delivered a driver-behaviour analysis system covering gaze, drowsiness, and occupancy, showcased at ETAS 2023
- Knowledge-distilled YOLOv7-E6E into YOLOv5-S for Jetson Nano deployment with only 8% mAP-0.5 loss

PROJECTS

Reinforcement Learning Zoo [GitHub]

Personal/FOSS Project

PyTorch, Gymnasium, MuJoCo, Atari/ALE, Weights & Biases

- Built an RL benchmarking hub of 9 algorithms spanning MLP, CNN, and GNN policies (PPO, SAC, TD3, DDPG, DQN, A2C, REINFORCE, SARSA, GNN-PPO) under one unified interface
- Verified results with seeded, reproducible evaluation matching paper-level scores — 5512 ± 25 on MuJoCo Humanoid-v5 and 100/100 on ALE/Boxing-v5

Tiny LLM [GitHub]

Personal/FOSS Project

PyTorch

- Built a 60M-parameter GPT-style LLM from scratch — BPE tokenizer, transformer with hand-rolled KV cache, and the full training stack (pretraining → instruction SFT → reasoning fine-tuning)
- Implemented tool-augmented reasoning via ChatML `<CALC>` tokens spliced with a safe AST calculator at decode time, plus bfloat16 mixed-precision multi-GPU DDP training and a stdlib-only token-streaming web chat

torchinfo [GitHub]

Open-Source Contributor

PyTorch

- Implemented accurate parameter and computation counting for modern architectures (LLMs, VLMs, YOLO detectors), fixing counts for tied embeddings, weight-shared submodules, and non-leaf-layer parameters
- Fixed kernel-size handling for 0-d tensors and expanded supported inputs to dictionaries with non-tensor values, scalar tensors, and `torch.distributions` outputs

ACHIEVEMENTS

- **OpenAI Codex for Open Source grant** (\$1,320) — August 2026
- **Kaggle Expert** — October 2024
- **High Achiever Scholarship** — May 2023

SKILLS

- **Programming:** Python, C/C++, CUDA, Rust, Bash **Frameworks:** PyTorch, TensorFlow
- **MLOps & Infrastructure:** Docker, AWS, Kafka, W&B (1,000+ hrs), Elasticsearch, ...
- **Spoken languages:** Vietnamese (native), English (IELTS 8.0), Chinese (HSK4)